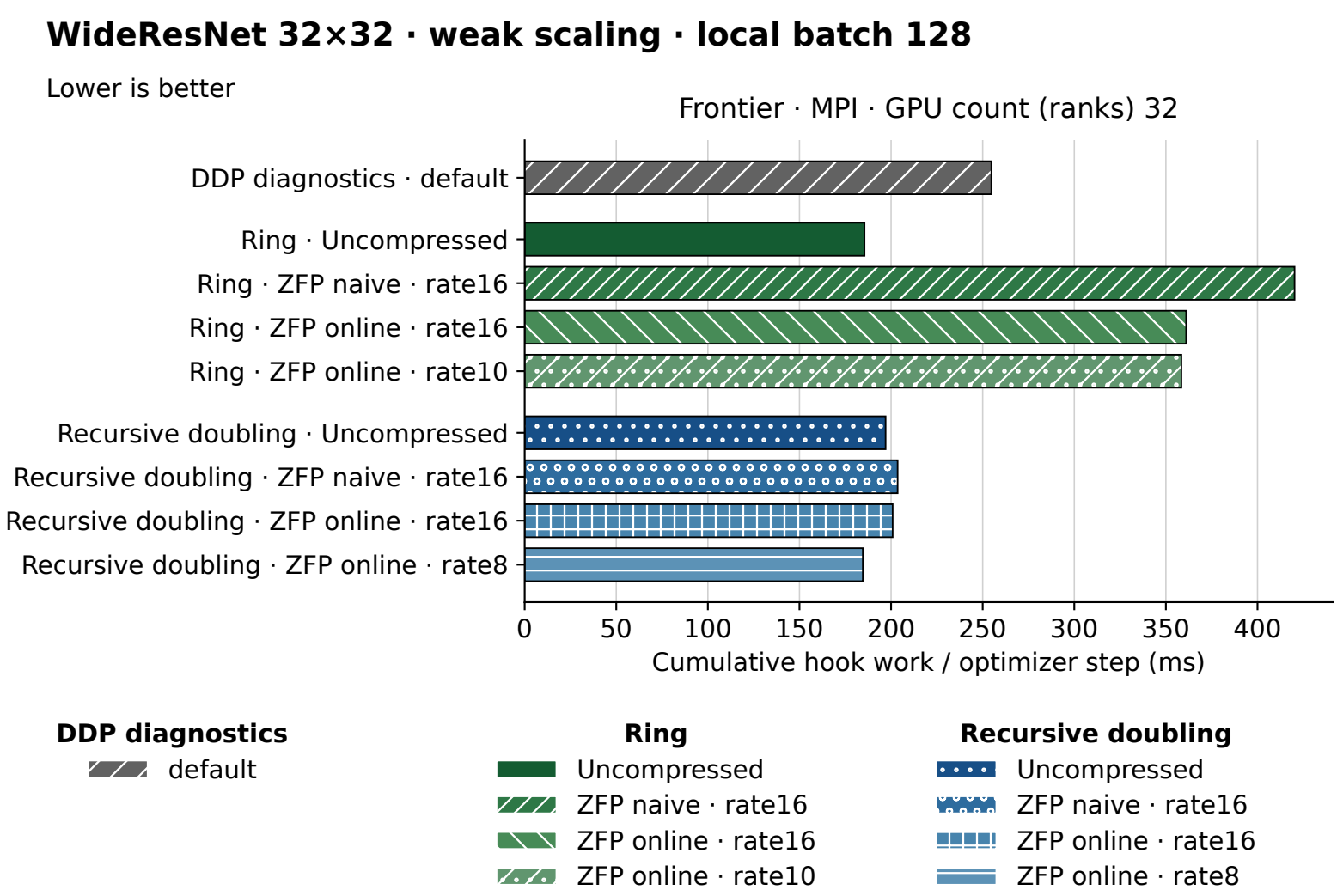

WideResNet 32×32 · weak scaling · local batch 128
Lower is better
Frontier · MPI · GPU count (ranks) 32
DDP diagnostics · default
Ring · Uncompressed
Ring · ZFP naive · rate16
Ring · ZFP online · rate16
Ring · ZFP online · rate10
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Recursive doubling · ZFP online · rate16
Recursive doubling · ZFP online · rate8
0
50
100
150
200
250
300
350
400
Cumulative hook work / optimizer step (ms)
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate